Exploring the Transcriptome: An Introduction to RNA-Seq Data Analysis

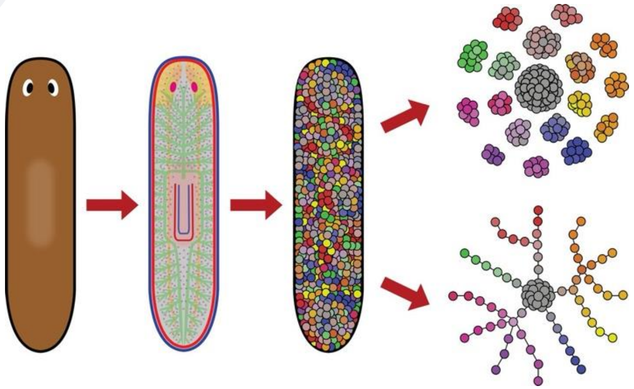
Presented by:
Arun Boddapati
Bioinformatics Scientist
SciComp, OAMD, NCEZID, CDC

Agenda for Today

- ❑What is transcriptomics and its role in public health & outbreak response ?
- ❑What are the components of transcriptomics and its applications ?
    1. When to use Bulk-RNA Seq vs Single cell RNA-Seq
- ❑Hybridization vs High-throughput?
- ❑Best Practices
- ❑Examples of GUI, Command line and R based tools
- ❑Current SciComp Projects

Transcriptomics

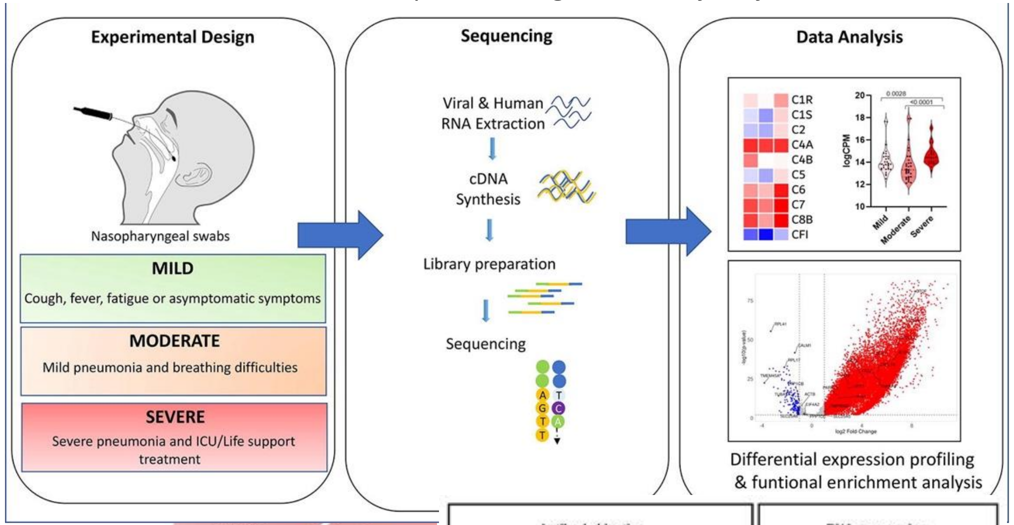
- Simple definition:
  - Study of RNA molecules present in a sample
- Complex definition:
  - Study of RNA molecules expressed from a genome at a particular point in time, space and condition.

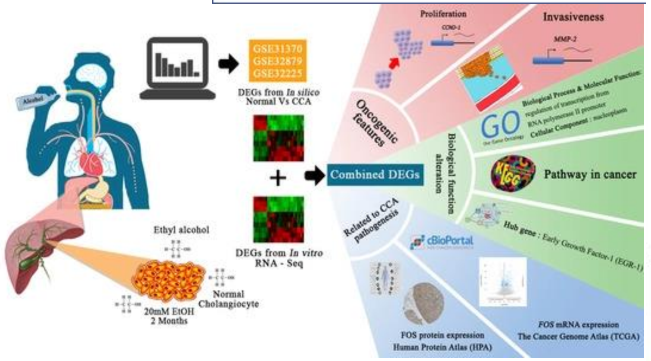


Transcriptomics in Public Health & Outbreak response

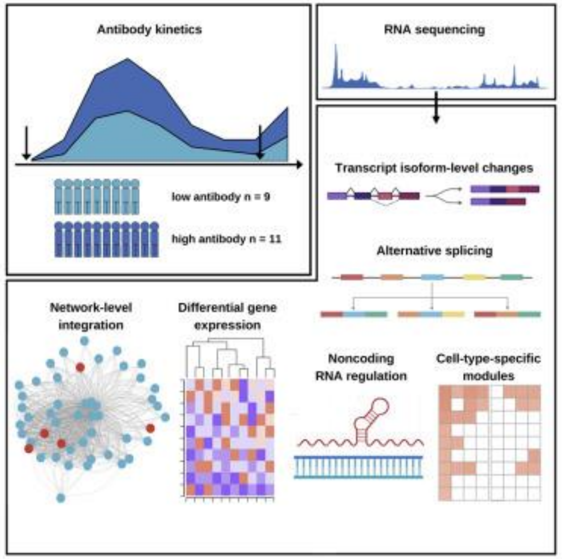
- Mechanisms of disease
- Understanding host response
- Identifying Biomarkers
- Identification of novel & emerging pathogens
  - E.g., Studying RNA viruses through RNA sequencing, because they cannot be studied via DNA sequencing.
  - Unless, the virus is a retrovirus which can integrate into host genome by using cDNA synthesis machinery
- Development of diagnostic tests & personal medicine
- Real-time monitoring of outbreaks

https://doi.org/10.1016/j.csbj.2020.12.016





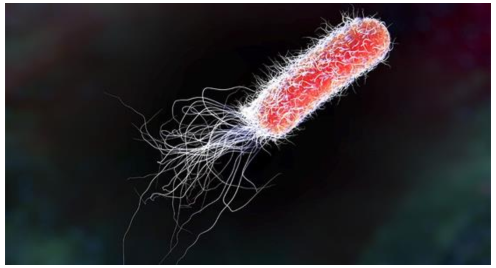
https://doi.org/10.3390/ijms20235987

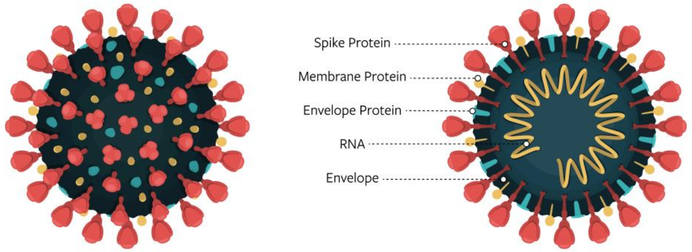


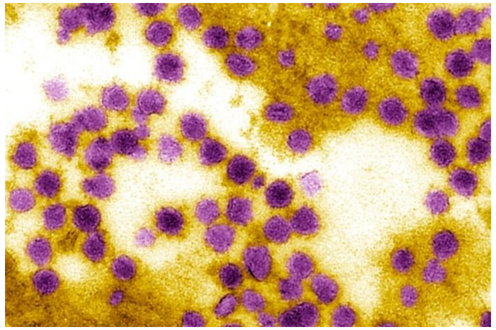
https://doi.org/10.1016/j.celrep.2022.110341

Case studies using RNA-Seq:

- E.*coli* outbreak in 2018:
  - Results: The strain had acquired a unique set of genes that allowed it to cause severe illness in humans.
- SARS-CoV-2 outbreak 2020:
  - Virus spread rapidly thereby acquiring new mutations and causing a global pandemic.
  - Pathogenesis
  - Immune response of virus
  - Vaccine discovery (moderns, Pfizer, J&J and drug repurposing (Baricitinib, Remdesivir)
- West Nile virus outbreak 2021:
  - Results: Viral undergoes significant changes in gene expression in response to host defenses. Upregulation & downregulation of genes involved in viral replication and host immune responses were identified.

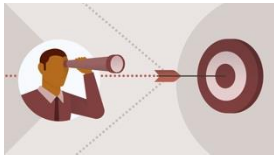


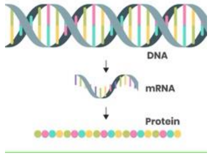




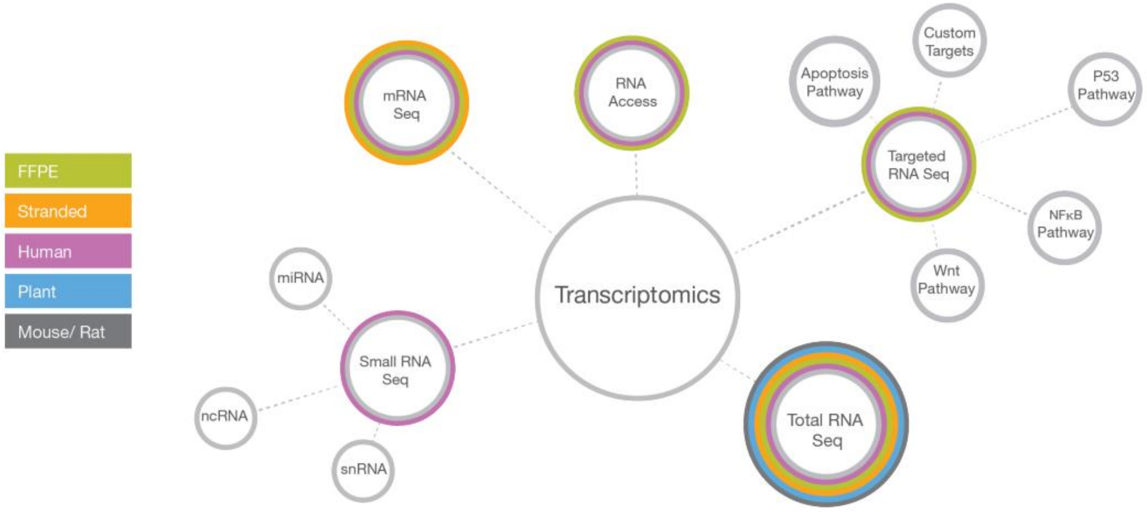
Components and applications

- mRNA, tRNA, rRNA, non-coding RNA, micro-RNA, small interfering RNA, long noncoding RNA in an organism is typically called a transcriptome.





- Messenger RNA (mRNA) code for proteins, that are perturbed by their gene expression changes associated with biological process or disease.
  - For example, mRNA sequencing can identify up & down regulated genes in response to specific treatment.



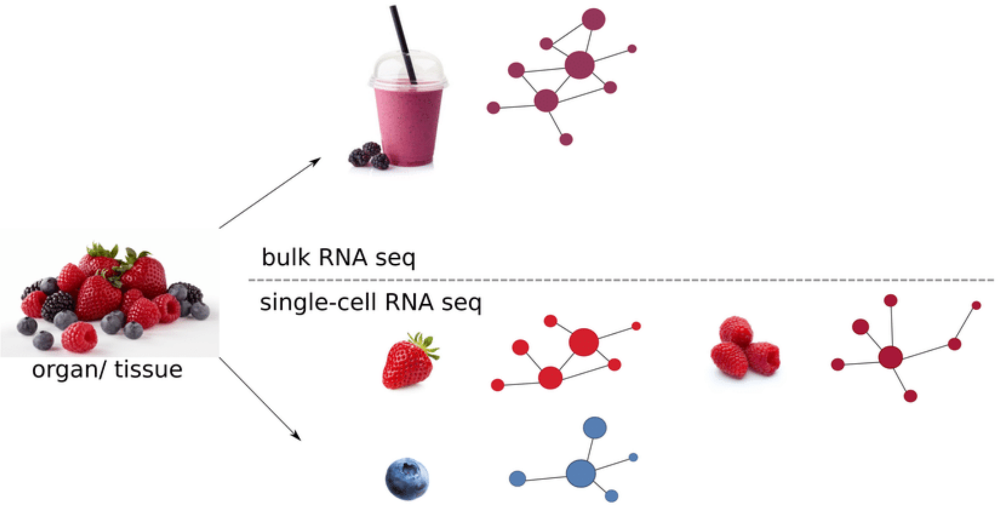
Applications:

1. Total RNA Sequencing
2. Targeted RNA Sequencing
3. Small RNA and Noncoding RNA Sequencing
4. Single-cell RNA Sequencing

- Transcriptome sequencing is a major advance in the study of gene expression because it allows a snapshot of the whole transcriptome rather than a predetermined subset of genes.
- It provides a comprehensive view of a cellular transcriptional profile at a given biological moment and enhances RNA discovery.
- Its unlimited dynamic range allows identification and quantification of both common and rare transcripts.

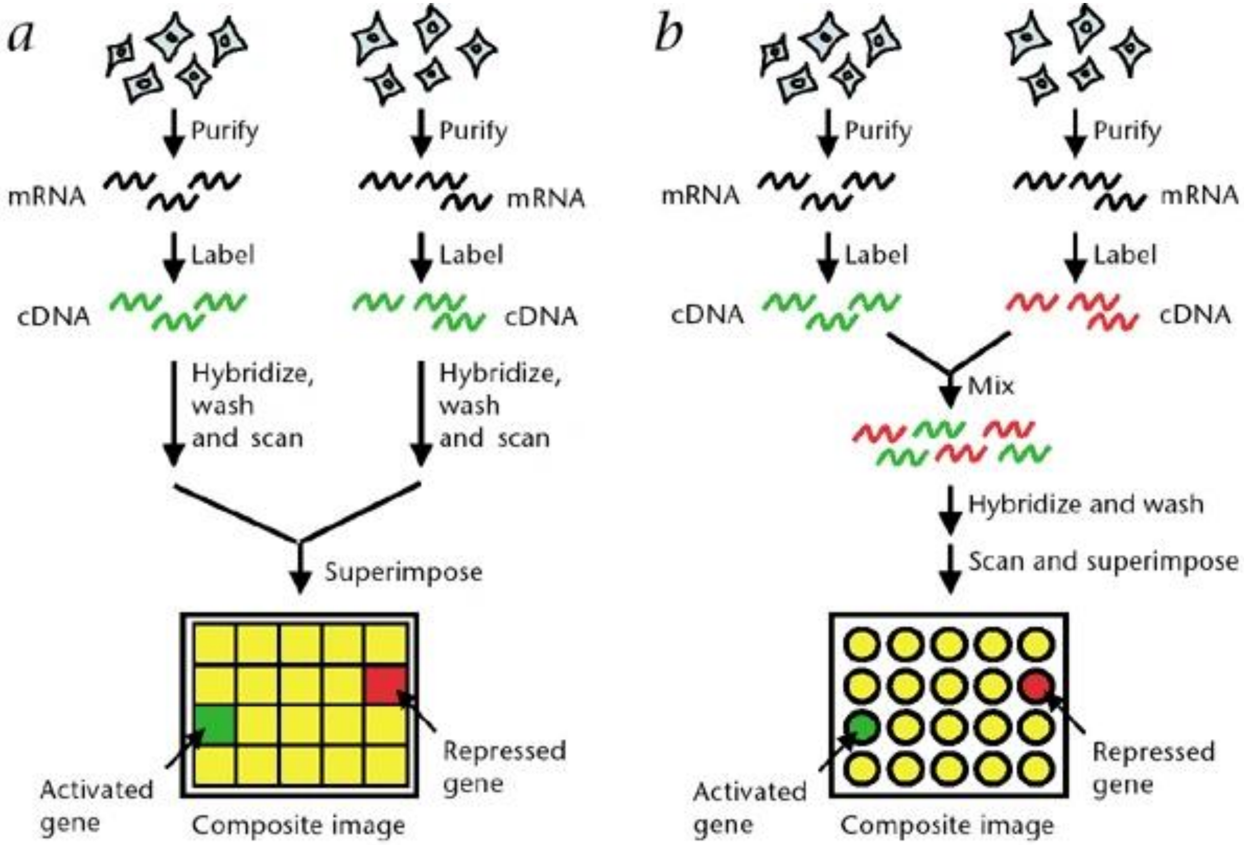
When to do Bulk vs Single cell RNA-Seq?

-  Cell-to-cell heterogeneity
-  Homogeneous tissue
-  High-resolution of gene expression
-  Different experimental conditions or tissue types
-  Rare cell populations
-  Broad overview of gene expression pattern
-  Heterogenous tissue or mixed population



Hybridization:

- Gene chips contain sequence probes of known genes from different conditions (normal & disease).
- when we add fragmented RNA, the sequence of genes are bound to the probes, and the intensities are scanned using confocal laser.
- A relative expression change is recorded.
- Either one color or two color arrays can be used as shown in the figure.



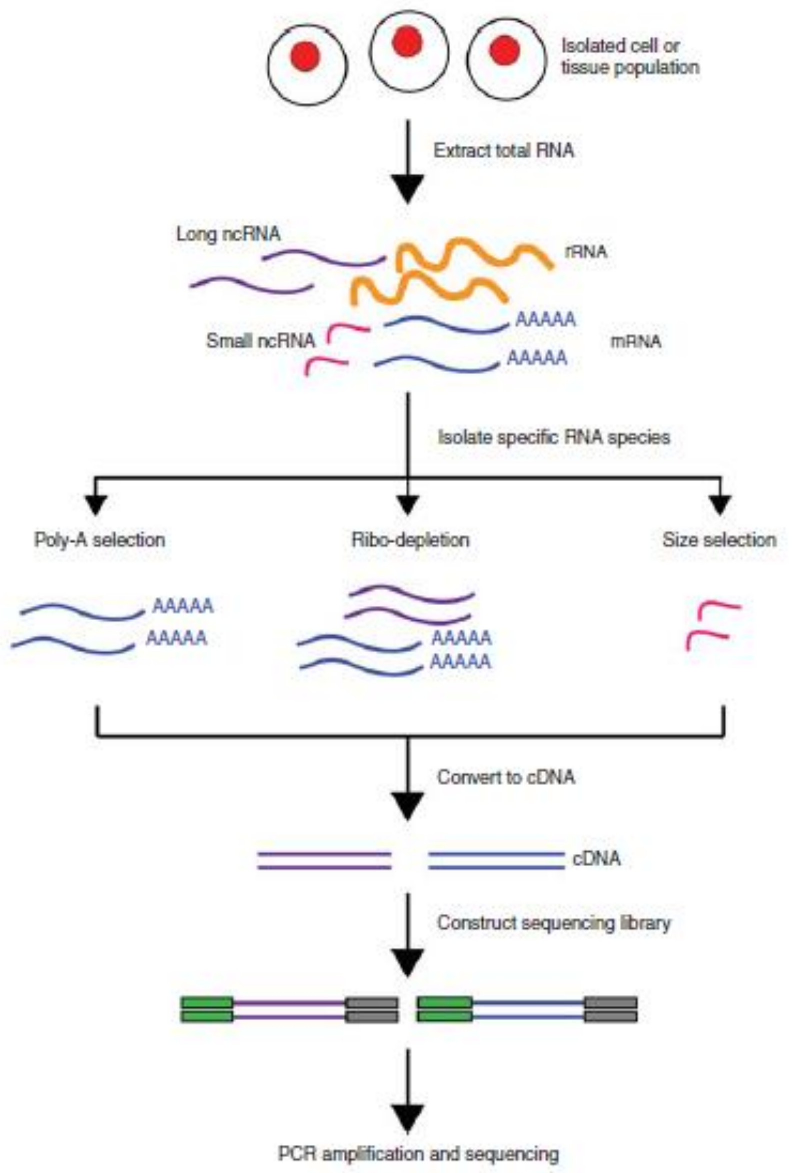
[Trends in microarray analysis | Nature Medicine]

High-throughput:

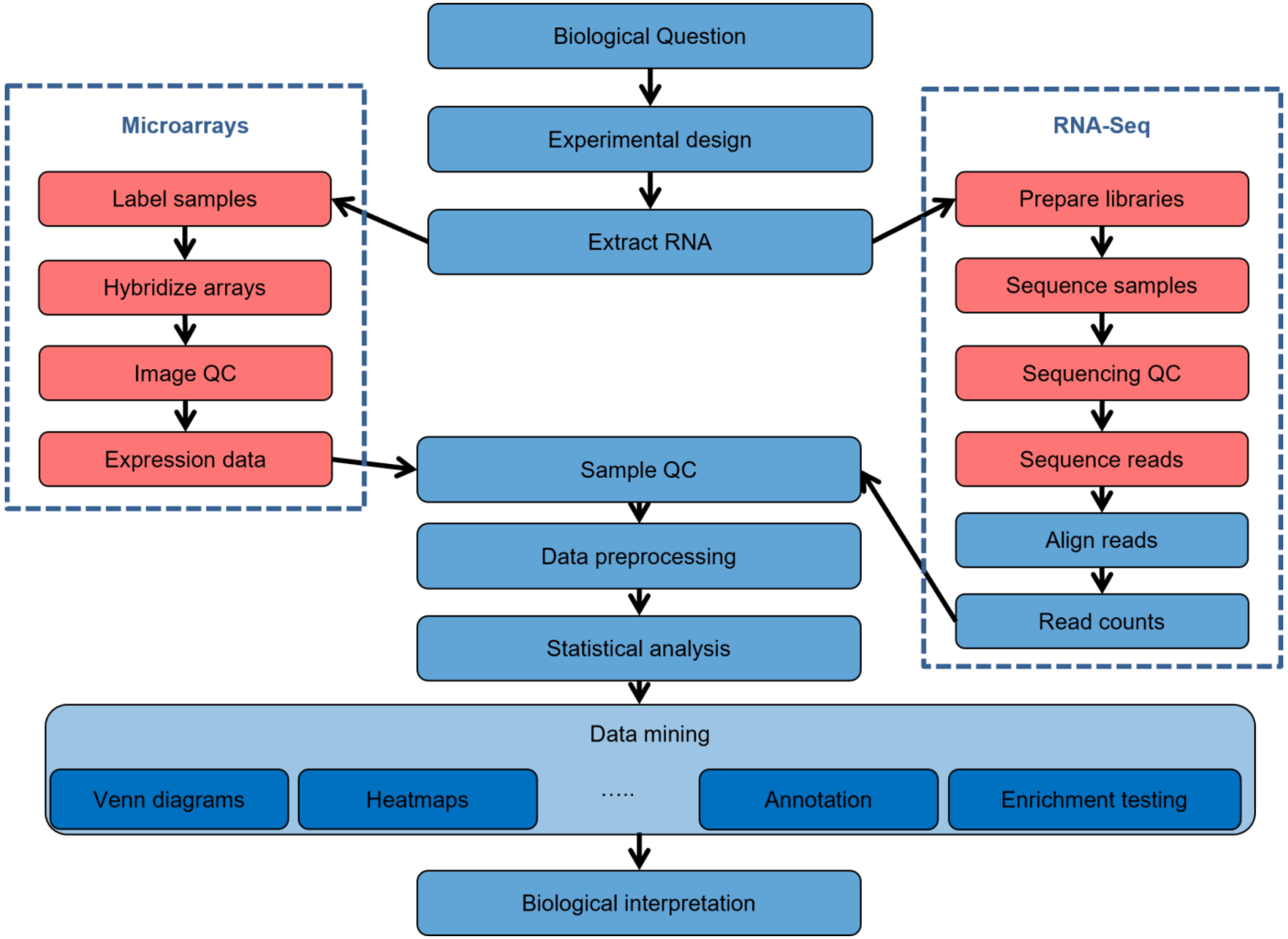
- RNA-Seq describes the abundance and sequence of RNA transcripts and its based on the usage of a reverse transcriptase to convert RNA to cDNA.
- Either oligo(dT) primers or random primers are used for this reaction.

Key differences:

  - oligo(dT) primers are highly 3' biased and mostly suitable for mRNA abundance estimations.
  - random primers are much suitable for total RNA abundance estimates.

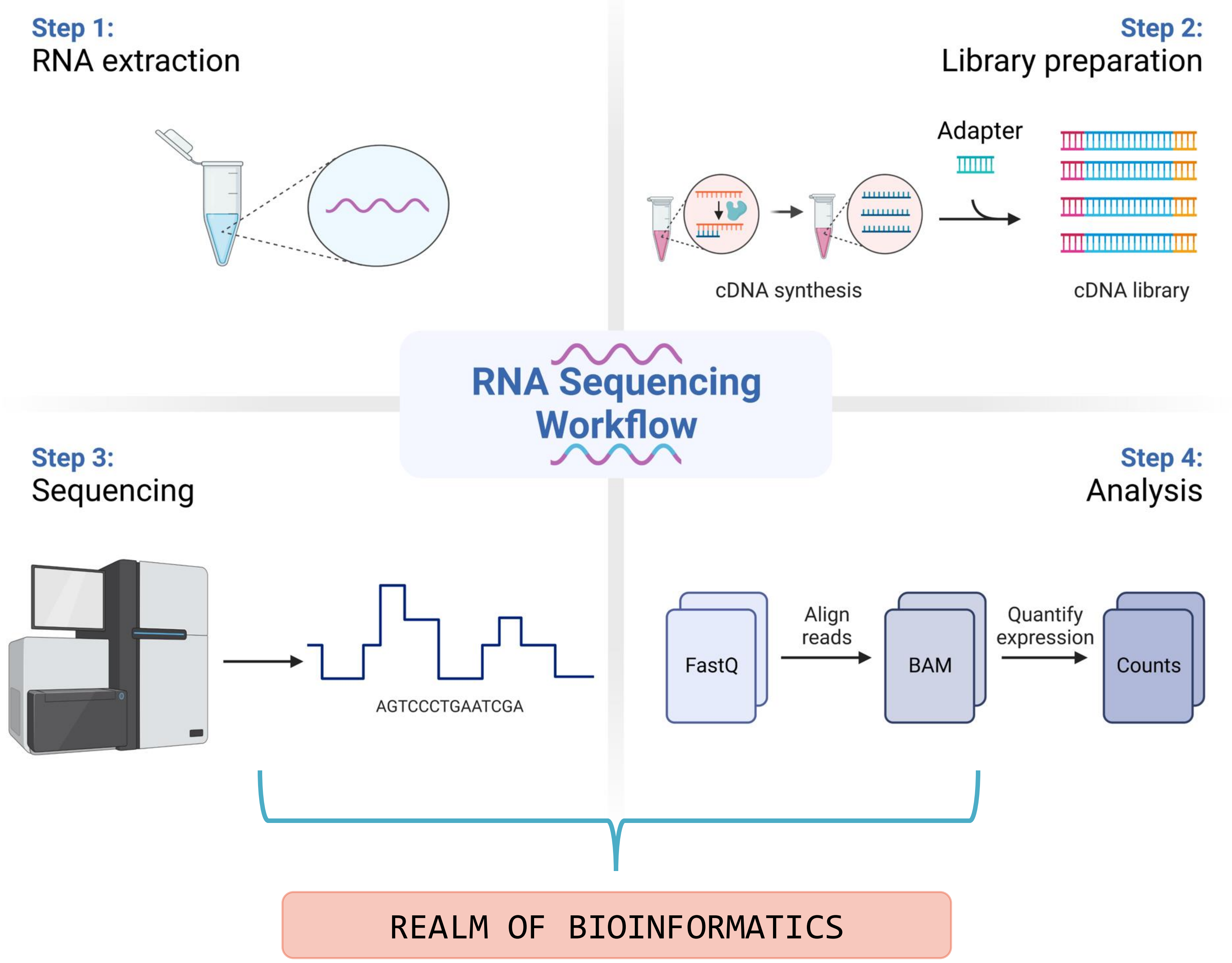


Hybridization vs High-Throughput Sequencing



The key differences between the two are:

- Microarrays are based on Hybridization probes that require prior sequence knowledge which doesn’t allow detection of novel transcripts.
- RNA-Seq ability to detect novel transcripts.
- Wider dynamic range
- Higher specificity and sensitivity, RNA-Seq can detect genes with low expression
- Simple detection of rare and low abundance transcripts, the sequencing depth can be tweaked to detect these transcripts.


Step 1:
RNA extraction
Step 2:
Library preparation
Adapter
cDNA synthesis
cDNA library
RNA Sequencing
Workflow
Step 3:
Sequencing
AGTCCCTGAATCGA
Step 4:
Analysis
FastQ
Align
reads
BAM
Quantify
expression
Counts
REALM OF BIOINFORMATICS

We have to address this:

Did you know there are Best practices for RNA-Seq?

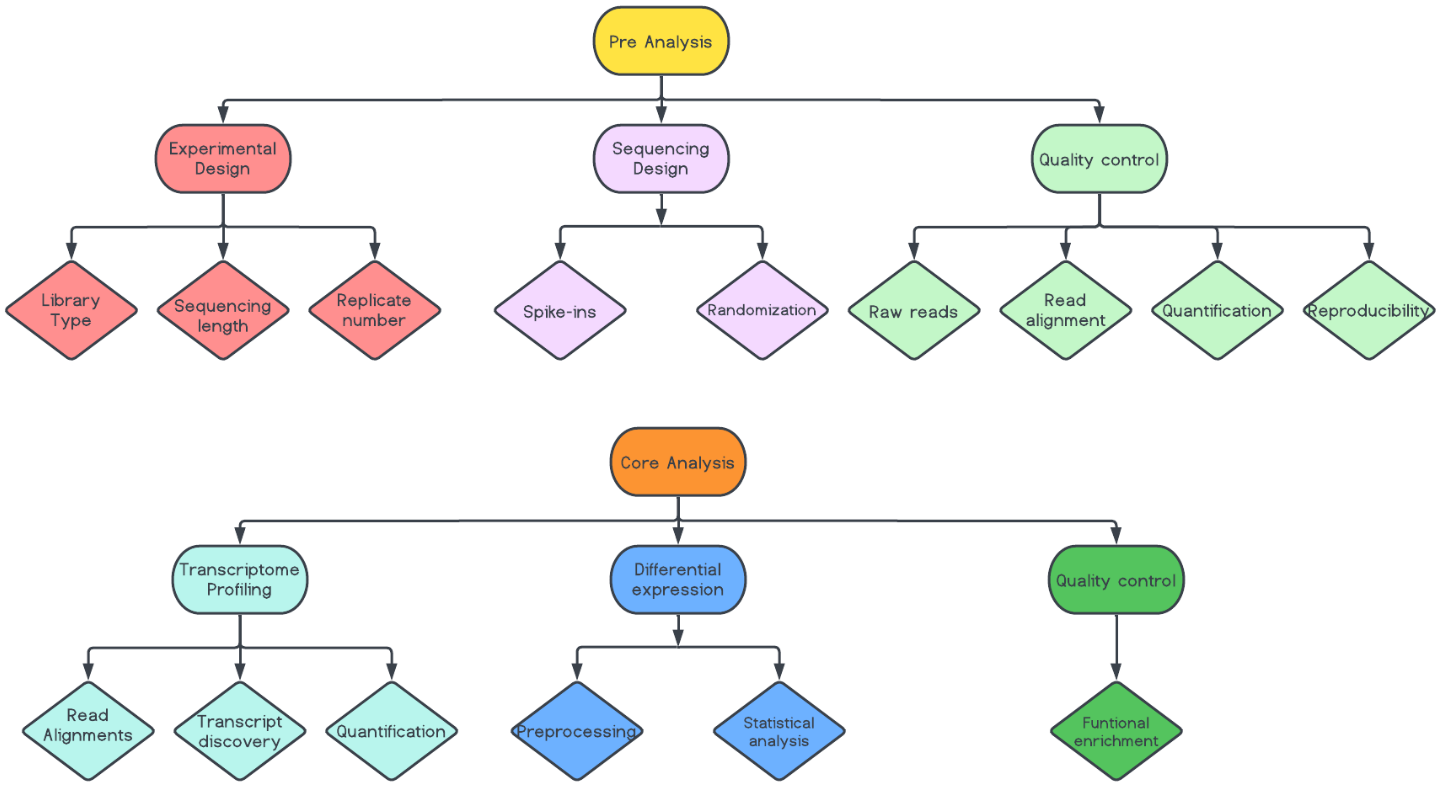
Pre-Analysis Steps:

Experimental Design:
- Library type
- Sequencing Length
- Replicate number

Sequencing Design:
- Spike-ins
- Randomization
  - Library prep**
  - Sequencing run**

Quality control
- Raw reads quality
- Read alignment
- Quantification
  - 3' bias
  - Low counts
- Reproducibility

BEST PRACTICES FOR RNA-SEQ ANALYSIS



Adapted from :
https://genomebiology.biomedcentral.com/articles/10.1186/s13059-016-0881-8

****Batch affects**

Core-Analysis:

Transcriptome profiling:
- Read alignment
- Transcript discovery
- Quantification level & measure

Differential expression:
- Preprocessing
- DEGs

Interpretation:
- Functional enrichment

- Ingenuity Pathway analysis (IPA)
- DNA Nexus**
- Partek Flow**

Graphic User Interface tools

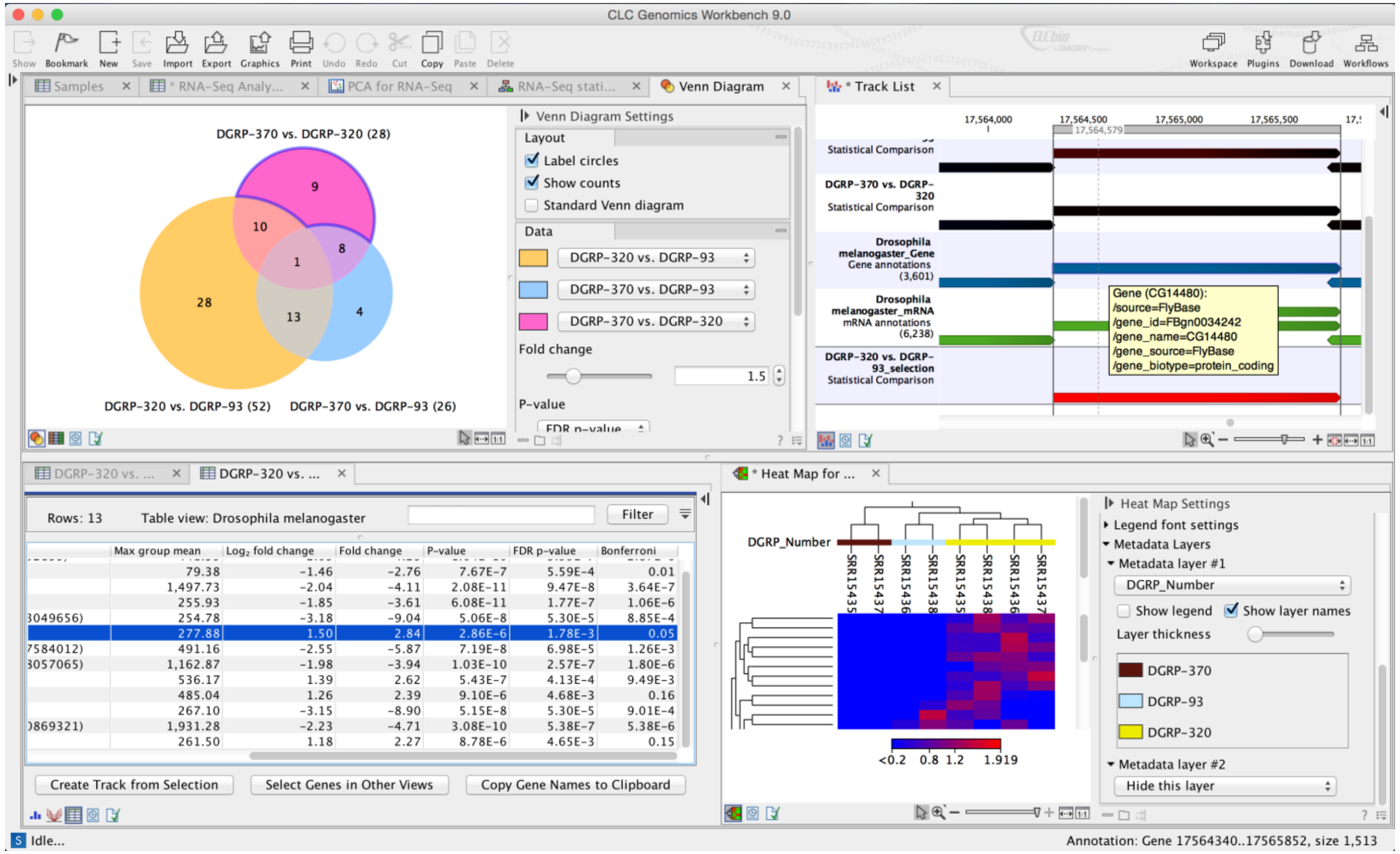
Advantages:

1. User-friendly
2. Plug & Play
3. Time saving
4. Visualization
5. **Doesn't require programming skills**

Disadvantages:

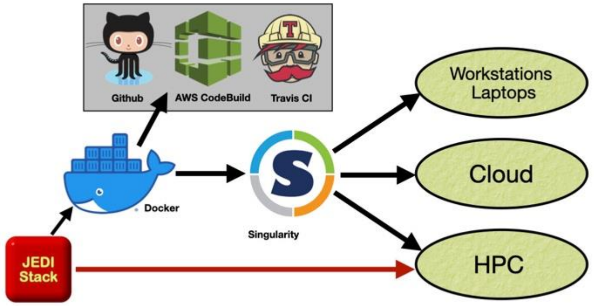
1. Costly
2. Cannot customize much
3. Limited flexibility
4. Efficiency

CLC Genomics Workbench**



https://info.biotech.cdc.gov/info/vendor-training-recordings/

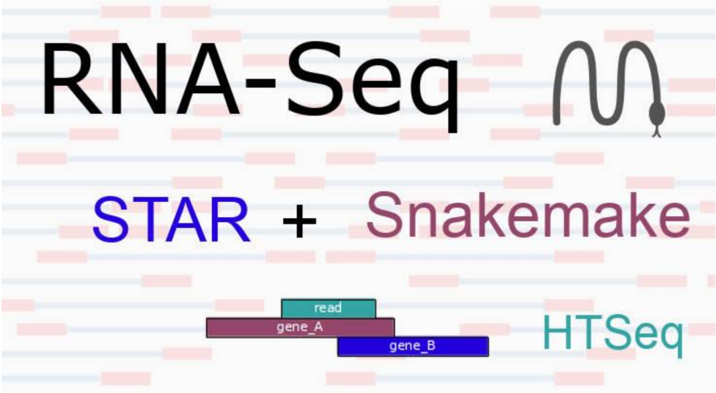
**SciComp supports these tools

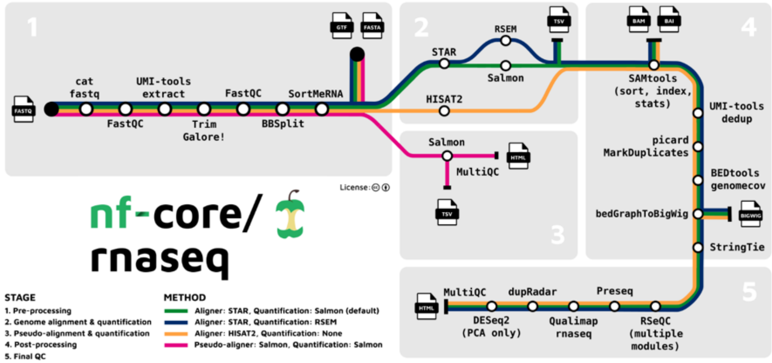


Containers**

Command Line tools (CLI)

Tools + Workflow Language





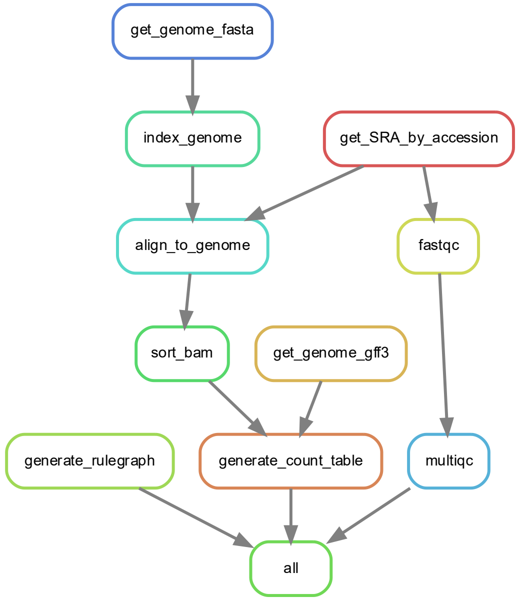
Nextflow**

Advantages:

1. Open-source (no cost)
2. Flexible
3. Reproducible
4. Efficiency
5. Highly customizable
6. Portable using docker containers

Disadvantages:

1. Learning curve
   - Require programming skills like R & Python
2. Debugging
3. No interactive Visualization



Snakemake

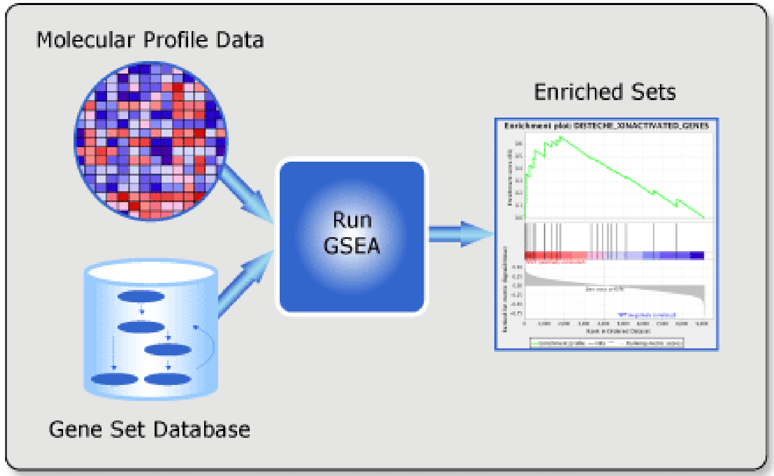
**SciComp supports these tools

R based tools for Analysis

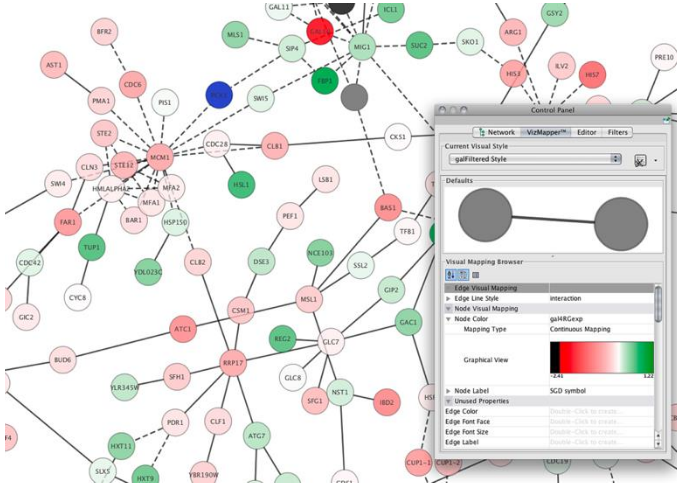
Downstream analysis tools for Differential expression

Visualization tool

Gene set enrichment tool for pathway enrichment



Network analysis and visualization



SciComp supports all tools